Worksheet 9.1

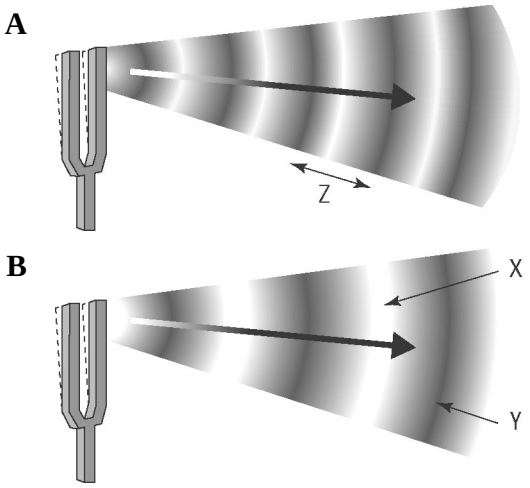
Science Quest 9: pages 308–11

Sound energy

Student: ..... Class: .....

Wavelength and frequency

1. The following diagram shows two vibrating tuning forks producing sound waves.



(a) Identify the features of the wave labelled X, Y and Z.

X: .....

Y: .....

Z: .....

(b) Which sound wave has the lowest pitch? Explain.

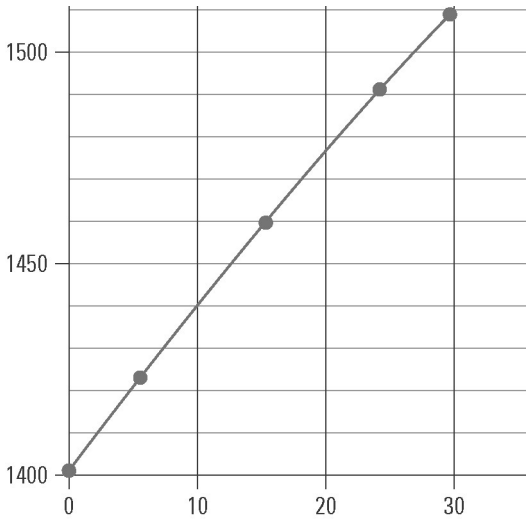
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Speed of sound in water

2. The following graph shows how the speed of sound (in m/s) in water varies with temperature (in °C).



(a) The labels and units on each axis are not shown. Insert these labels and units.

(b) Give the graph a title.

(c) Use the graph to determine the speed of sound in water at:

(i) 0 °C .....

(ii) 10 °C .....

(iii) 20 °C .....

(d) In air, the speed of sound at 0 °C is 331 m/s.

(i) How many times faster does sound move in water than air at that temperature? (*Hint:* Divide one speed by the other.)

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(ii) Explain why sound moves faster in water than in air.

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